Anshuman Das

Bengaluru, India | +91 72629 22007 | dasanshuneel001@gmail.com
linkedin.com/in/anshuman-das-875a9a282 | github.com/AnshumanDas001

Summary

AI and backend engineer focused on production LLM systems: Retrieval-Augmented Generation (RAG), model serving, safety guardrails, and cost-aware LLM routing, built with Python, FastAPI, PostgreSQL, and Docker.

Experience

ACT Fibernet Jan 2026 – Present
Management Trainee, Technology Bengaluru, India

- Architected a **production RAG platform from scratch** for Network Operations Center (NOC) workflows, serving **200 engineers**: **MinerU Pro** document parsing, **Gemma 4 vision** image descriptions, **Qwen3-4B** embeddings, and **Qwen3.6-35B** generation.
- Optimized LLM inference with **vLLM and llama.cpp**, cutting **GPU VRAM usage by 40%** and scaling concurrent users **10x** (**2 to 20**).
- Built a **two-layer guardrails service**: **Llama Guard** for safety filtering and a **fine-tuned DistilBERT** classifier for domain filtering, rejecting unsafe and off-domain queries.
- Designed separate **authentication and admin microservices** for user and document management, enforcing **role-based access control (RBAC)** at the service level and per vector database collection.

Technologies: Python, PostgreSQL, vLLM, llama.cpp, MinerU, Qwen3, Gemma, Llama Guard, DistilBERT

Texas Instruments May 2025 – Jul 2025
Summer Intern

- Built a **Python/Jenkins continuous integration (CI) pipeline** automating LabVIEW/TestStand hardware validation.
- Optimized test suites with Python unit tests and shell scripts, cutting test turnaround time by **66%** (**3x faster** feedback).

Technologies: Python, Jenkins, Shell Scripting, LabVIEW, NI TestStand

Projects

ThriftLLM: Cost-Optimized LLM Router with Verified Escalation Live Demo | GitHub

- Built and deployed an **OpenAI-compatible LLM gateway** with multi-user auth, cost tracking, and encrypted key storage; on a **116-query**, three-difficulty benchmark it cut inference cost **82%** vs. a frontier model and **41%** vs. a mid-tier model, with all **76 objectively graded answers correct**.
- Designed a routing policy that minimizes **expected cost** (generation, verification, and escalation) under a **0.80** per-difficulty accuracy floor; the routing map is derived **automatically** from calibrating any user-supplied model.
- Classified prompt difficulty in about **7 ms** via a **similarity-weighted k-NN** vote over sentence embeddings, raising accuracy from **56.0% to 64.7%** with no LLM call.
- Built **two-stage verification**: a logistic-regression confidence gate over token logprobs, then a **$0.00004** LLM judge that catches **94%** of incorrect cheap-tier answers before escalation.

Technologies: Python, FastAPI, LiteLLM, Sentence-Transformers, scikit-learn, Turso (libSQL), Google Cloud Run

Event Plan Optima: Event-Driven Traffic Congestion Forecasting Live Demo | GitHub

- Built a congestion forecasting dashboard for **Bengaluru Traffic Police** from **8,173 real event records**, recommending **manpower, barricading, and diversion** plans for planned and unplanned events.
- Trained a **logistic-regression road-closure model** (**0.77** CV AUC, on par with Random Forest) after rejecting a **leaky target**, and ported it to JavaScript for **zero-backend, interpretable inference**.

Technologies: Python, pandas, scikit-learn, JavaScript, Leaflet, Chart.js, Netlify

Education

Bachelor of Technology in Electronics and Instrumentation Engineering Nov 2022 – May 2026
National Institute of Technology, Agartala **CGPA: 8.04/10**

Skills

Languages: Python, C++, C, JavaScript, SQL
AI and LLM: RAG, LLM Serving, vLLM, llama.cpp, LLM Routing, LLM Evaluation, Guardrails, Fine-Tuning, Vector DBs
Backend: FastAPI, Flask, REST APIs, Microservices, PostgreSQL, Redis, SQLite
Cloud and DevOps: Docker, Jenkins, CI/CD, Google Cloud Run, Linux, Git, Shell Scripting
Networking: TCP/IP, OSI Model, OSPF, RIP, Packet Analysis, WebSockets
CS Fundamentals: Data Structures and Algorithms, Operating Systems, DBMS, OOP, Design Patterns

Achievements

- **LeetCode Knight** with a rating of **1898** (**top 4.42%** globally); solved **600+** problems on LeetCode and GeeksforGeeks.
- **Top 5%** of **30,000+** applicants in **Flipkart GRiD 8**; **Special Mention** at Model United Nations.